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RESEARCH ARTICLE

Prediction and Explainability of School Dropout in Peru Using Machine Learning and Explainable Artificial Intelligence

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ABSTRACT School dropout represents a critical challenge for educational systems in developing countries. This study proposes an integral framework that combines machine learning techniques and explainable artificial intelligence to predict and explain school dropout in the Peruvian context. The proposal incorporates key variables in four dimensions: personal, family, sociodemographic, and academic. The study implements and compares five machine learning algorithms (Random Forest, Support Vector Machine, Gradient Boosting, K-Nearest Neighbors, and XGBoost) along with SHapley Additive exPlanations (SHAP) for explainability, integrated into a web application called EduPredict. The experiments conducted on a dataset of 1,456 records from a public school in Metropolitan Lima, using 20 predictive factors, demonstrate exceptional performance, with Gradient Boosting achieving an accuracy of 97.27% and precision of 99.28%. The explainability analysis using SHAP identified behavior and academic performance as the most determining factors, significantly contributing to the dropout risk. The validation with 10 educational professionals through structured surveys revealed high levels of usability, utility, and satisfaction, confirming the practical applicability of the system in real educational environments. The framework has potential to be adapted to other educational contexts in developing countries, offering a personalized approach to mitigate school dropout.

INDEX TERMS School dropout, machine learning, explainable artificial intelligence, SHAP, educational predictors, dropout prediction.

I. INTRODUCTION

School dropout, defined as the abandonment of studies before completing an educational cycle [1], constitutes a critical challenge for global socioeconomic development. Its causes are multifactorial and include economic, social, cultural, and personal aspects, as well as deficiencies in educational support and student motivation. According to recent data [2], around 250 million children and young

people—approximately 16% of the school-age population worldwide—are not enrolled in educational institutions. This figure reflects an increase of 6 million since 2021, signaling a setback in global educational advances. The situation is especially critical in regions like Afghanistan, where 7.8 million children lack access to education, with enrollment rates of just 50% in primary and 20% in secondary [2]. If urgent interventions are not implemented, this trend threatens to deepen educational gaps, putting at risk the achievement of the Sustainable Development Goals (SDGs) for 2030. In Latin America and the Caribbean, while coverage

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rates in 2020 reached 93.2% in lower secondary and 78.7% in upper secondary, the COVID-19 pandemic exacerbated school dropout, according to [3]. In Peru, the national school dropout rate reached 1.36% in 2021, representing more than 110,138 students who interrupted their studies [4].

The problem of predicting school dropout consists of anticipating—through the analysis of historical data—which students present a higher risk of leaving the educational system. Complementarily, explainability allows interpreting the reasons behind these predictions, identifying not only the risk factors but also their relative influence on the student's decision [5]. This duality (accuracy + interpretability) is fundamental for designing effective interventions, such as personalized tutoring, psychosocial support, or curricular adjustments, that mitigate dropout and improve school retention.

Recently, machine learning (ML) has shown significant advances in this field. ML is a subset of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without explicit programming for each task, making it particularly useful for analyzing complex educational datasets to identify at-risk students [6]. Studies like [7] achieved 85% accuracy in identifying at-risk students in Egypt using a logistic classifier for a dataset of 3,154 students. In Brazil, [8] reached 84% accuracy with neural networks using a sample of 3,200 students, while [9] obtained 88% with XGBoost in online learning environments for 1,500 students. Other innovative approaches include the work of [10] in Germany, which combined sociodemographic data with digital interactions achieving 78% accuracy for 7,000 students, and the model of [11] in Chile, which used LightGBM obtaining 99.7% accuracy for 4,500 students. Notably, the use of SHAP by [12] in Brazil allowed explaining the most influential variables in predictions with data from 13,931 students. In Peru, although no studies on the prediction and explanation of dropout have been identified in the literature, the national approach is limited to reactive protocols, such as home visits after dropout has occurred, instead of adopting proactive methods to anticipate it [13].

Current studies employ ML techniques to predict school dropout in various educational contexts. However, these models present limitations in accuracy, with error rates exceeding 10% [11], [12], which reduces the effectiveness of interventions in at-risk students. Additionally, these models cannot be extrapolated to other contexts like the Peruvian one because they contemplate different realities that manifest in their factors and data; for example, travel time, not considered in several studies, is relevant in rural areas where children walk long hours to reach school, a factor closely related to academic performance and consequently to permanence in the educational system. Therefore, it is necessary to propose a prediction and explainability model for Peruvian school dropout.

The integration of explainable artificial intelligence (XAI) techniques in predictive models of school dropout allows

interpreting the bases of the model's decisions, identifying critical factors and quantifying their influence on the risk of student dropout. These advances enable designing evidence-based and highly personalized retention strategies. Despite progress in the field, a critical research gap remains: no prior study has simultaneously addressed prediction accuracy, model explainability, and practical deployment in the specific Peruvian educational context. Existing approaches either lack XAI integration, focus on higher education rather than secondary-level dropout, or fail to account for context-specific factors critical in developing countries (e.g., travel time to school and family socioeconomic conditions in peri-urban Lima). This work directly addresses this gap by proposing a framework validated in an authentic Peruvian school setting.

This study develops a web application that combines ML with XAI to predict and explain school dropout in the Peruvian context. The implementation of explanatory methods like SHAP enables not only achieving high predictive accuracy but also determining the specific weight of various academic, socio-family, and personal risk factors. The resulting platform strengthens institutional capacities to implement early and evidence-based interventions. The main contributions of this work are:

- A predictive and explanatory model of school dropout tailored to the Peruvian educational context, combining five ML algorithms with SHAP-based XAI;
- EduPredict, a web application implementing the proposed model for real-time dropout risk prediction and individual-level explainability;
- A case study validating the framework on 1,456 student records from a public secondary school in Metropolitan Lima, demonstrating high predictive performance and practical applicability confirmed by 10 educational professionals.

The article is structured as follows. In Section II, a literature review on the prediction of school dropout is presented. The proposed predictive explanatory model is described in Section III, and its implementation in Section IV. The validation of the model and the application and its discussion follow in Sections V and VI respectively. Finally, the conclusions follow in Section VII.

II. LITERATURE REVIEW

Studies related to risk factors, prediction, and explainability of school dropout in the context of ML and XAI are reviewed.

A. RISK FACTORS

School dropout is a multifactorial phenomenon influenced by personal, family, sociodemographic, and academic dimensions. Personal factors include individual characteristics such as gender, age, and health, which affect student performance and permanence [14]. Family factors encompass home environment conditions that affect the educational trajectory [7], while sociodemographic factors reflect socioeconomic and

geographic disparities in the student's context [10]. On the other hand, academic factors, such as performance and attendance, directly influence learning [15]. Their effect is usually cumulative, exacerbating the risk when they coexist [7]. The early identification of these factors not only allows preventive interventions but also underpins predictive and explanatory models of dropout. Given the large number of studies on school dropout factors, state-of-the-art studies have been developed. [12] identify 137 factors and categorize them into dimensions such as educational stage, student information, extracurricular classes, economic situation, fees, parent information, location statistics, current grades, cumulative grades, grade difference from the previous quarter, grade difference from the previous year, and dropout. Other studies, such as [16], highlight 29 factors and categorize them into dimensions such as enrollment data, performance, and scholarships. In Table 1, 50 school dropout risk factors are presented, categorized by four dimensions (personal, family, sociodemographic, and academic) according to [17].

TABLE 1. School Dropout Risk Factors and Their Classification into Four Dimensions. Each factor is supported by at least one empirical study; references are shown in the third column.

Dimension	Factor	Reference
Personal	Age; Marital status; Health condition; Ethnic affiliation; Extracurricular activities; Number of siblings; Level of self-concept; Reading and mathematics skills; Self-esteem; Emotional symptoms; Future aspirations; Peer support; Relationships with teachers; School motivation; Behavioral problems	[14], [17], [12], [7], [18], [19]
Family	Dysfunctional family structure; Parental relationship; Parental support; Parental educational level; Parental employment; Family income; Housing condition; Family size; Socioeconomic level	[20], [16], [15], [7]
Sociodemographic	Culture; Type of residential area; Access to basic services; Living conditions; Additional academic support; Social activity; Parents' working conditions; Community educational level; Social gap index; Employment situation	[17], [21], [15], [10], [22]
Academic	Academic performance; Classroom participation; Grades; Absenteeism; Grade repetition; Academic motivation; Educational resources; Student-regulated activity; Professional development activities; Admission examination scores; Enrollment history; Participation in tutoring programs	[14], [16], [7], [18], [20]

B. PREDICTION OF SCHOOL DROPOUT

A predictive model is a computational system that estimates the probability of future events through the analysis of

patterns in historical data [23]. Its essential components include: (a) machine learning algorithms, such as decision trees and artificial neural networks [6]; (b) training and validation data, where the model is trained using stratified cross-validation (10 iterations, 80%–20% training-test split) to prevent data leakage [7]; and (c) evaluation metrics, such as accuracy, precision, sensitivity (recall), specificity, and F1-score, to quantify the classifier's performance [7].

Table 2 describes 13 representative studies on school dropout prediction based on ML. These studies span multiple countries and education levels, illustrating both the diversity of approaches and the variability in reported performance.

It is worth noting that the literature covers other educational areas; for example, [27] analyze 67 studies in higher education, identifying 50 algorithms and 112 predictive factors, while [28] review 73 works and document 53 algorithms.

C. EXPLAINABILITY OF SCHOOL DROPOUT

Explainable artificial intelligence consists of methods and processes that help users understand the results generated by the predictive model [29]. Its main components are: (a) a machine learning algorithm, such as logistic regression and gradient-boosted trees [6], to predict an outcome; and (b) an explainability technique, which shows how factors contribute to the positive or negative class in the prediction [30]. Recent applications of XAI with SHAP have demonstrated high effectiveness across diverse domains, including IoT security [31], environmental monitoring [32], and natural language processing [33], validating SHAP as a robust and domain-agnostic interpretability framework. These precedents support its adoption in educational dropout prediction, where transparency is essential for practitioner trust and actionable intervention.

In Table 3, seven recent studies on explainability in school dropout are summarized, highlighting the prediction and explainability techniques employed, as well as the factors identified as most influential.

D. COMPARISON WITH EXISTING APPROACHES AND RESEARCH GAP

Table 4 summarizes the key differences between this work and the most closely related studies. Several observations emerge. First, the majority of existing ML-based dropout studies focus on higher education [12], [16], [17], whereas secondary-level dropout—particularly in developing countries—remains underexplored. Second, few studies combine high-accuracy prediction with deployed XAI tools accessible to non-expert educators; most provide SHAP analyses as offline research artifacts. Third, Peruvian-specific socioeconomic and geographic factors (e.g., travel time in peri-urban areas, multi-generational households) are absent from all reviewed studies. This work addresses these gaps by: (1) focusing on secondary-level dropout in Peru; (2) deploying a functional web application integrating ML

TABLE 2. Studies on School Dropout Prediction. Results highlight the diversity of contexts, methods, and performance metrics reported in the literature.

#	Ref.	Data Set	Method	Results
1	[18]	165,715 students (Korea)	RF	A: 95.0%, R: 85.0%
2	[24]	2,097 students, EPN (Ecuador)	LR; ANN	A: 72.7%, AUC: 47.5%; A: 76.8%, AUC: 79.5%
3	[16]	5,426 students (Spain)	SVM; GBM; RF; Ensemble	A: 90.8%, P: 89.0%; A: 88.4%, P: 87.0%; A: 88.4%, P: 88.3%; A: 88.8%, P: 86.4%
4	[25]	DS1: 78,623 (USA); DS2: 120,542 KDD (China)	CONV-LSTM	DS1: P: 91.0%, R: 88.0%; DS2: P: 93.0%, R: 87.0%
5	[20]	5,951 students, UAI & U Talca (Chile)	GBDT	R: 56.0%, A: 71.0%
6	[26]	79,186 secondary students (China)	MFCN-VIB; CNN; SVM; LDA; NB; CART	P: 88.7%, R: 96.0%; P: 88.2%, R: 95.8%; P: 88.3%, R: 95.2%; P: 85.9%, R: 97.4%; P: 88.1%, R: 92.0%; P: 87.7%, R: 88.4%
7	[7]	3,154 respondents, SYPE Survey (Egypt)	LR	F: 52.0%, AUC: 78.2%
8	[15]	2,668 primary students (Malawi)	RF; Lasso; Ridge; MNN	R: 73.4%, P: 31.5%; R: 80.6%, P: 31.7%; R: 77.2%, P: 31.6%; R: 80.2%, P: 32.1%
9	[17]	1,847 students (Mexico)	DTs; ANN; LR	A: 93.4%, P: 99.5%; A: 93.3%, P: 97.7%; A: 93.3%, P: 99.6%
10	[22]	200,000 high school students (Turkey)	SVM; LR; K-NN; DTs; NB; RF	A: 94.8%, P: 96.8%; A: 93.3%, P: 94.2%; A: 93.1%, P: 95.2%; A: 93.5%, P: 95.0%; A: 93.9%, P: 94.0%; A: 91.0%, P: 95.9%
11	[19]	4,424 students (USA)	LSTM-DNN	A: 92.0%, P: 95.0%
12	[14]	4,160 students, School Path (Finland)	B-RF	A: 61.0%, R: 60.0%
13	[21]	1,080,782 respondents (Mexico)	ANN; LRALR; BO; SVM; RF	P: 91.0%

LR: Logistic Regression; RF: Random Forest; CART: Classification and Regression Trees; DTs: Decision Trees; ANN: Artificial Neural Network; GBDT: Gradient-boosted trees; LRALR: Linear Ridge and Lasso Regression; BO: Bayesian Optimization; SVM: Support Vector Machines; CONV-LSTM: Convolutional Neural Networks and Long Short-Term Memory; LSTM: Long Short-Term Memory; DNN: Deep Neural Network; KNN: K-Nearest Neighbors; NB: Naive Bayes; CNN: Convolutional Neural Networks; LDA: Linear Discriminant Analysis; GBM: Gradient Boosting; MNN: Multilayer Neural Network. Metrics: P: Precision; A: Accuracy; R: Recall; S: Specificity; F: F1 Score.

TABLE 3. Studies on School Dropout Explainability. SHAP is the dominant technique, consistently identifying academic performance and behavioral factors as the most influential predictors.

#	Ref.	Prediction	Explainability	Results
1	[30]	RF	SHAP	Lower average values, lower final trajectory values and higher bias values
2	[34]	MLP	SHAP; LIME	School average, minimum wages earned by the family, work characteristics of the student's mother
3	[12]	XGBoost	SHAP	Cumulative Portuguese, Cumulative Geography, Average grade
4	[11]	DTs; XGBoost; LightGBM; CatBoost	SHAP	Last approved qualification, number of repetitions, last grade complete
5	[35]	XGBoost; AdaBoost	SHAP; SAGE	Curricular units in the 2nd semester, tuition fees, curricular units in the 1st semester
6	[36]	RF; FTT	SHAP	Number of ECTS credits, weighted average grades, context of enrollment
7	[37]	LR; LDA; XGBoost; NGBoost; LightGBM; AdaBoost; CatBoost	GBC; PI; SHAP; LIME; PDP	GPA and mathematics exam scores; Grades in specific subjects and extracurricular activities; Math maturity test scores, high school GPA

SHAP: Shapley Additive Explanations; LIME: Local Interpretable Model-agnostic Explanations; PI: Permutation importance; PDP: Partial dependence plot; SAGE: Shapley Additive Global Importance. XGBoost: Extreme Gradient Boosting; LightGBM: Light Gradient Boosting Machine; MLP: Multilayer Perceptron; CatBoost: Categorical Boosting; NGBoost: Natural Gradient Boosting; LDA: Linear Discriminant Analysis; LR: Logistic Regression; RF: Random Forest; AdaBoost: Adaptive Boosting; DTs: Decision Trees; FTT: Feature Tokenizer Transformer; GBC: Gradient Boosting Classifier.

TABLE 4. Comparison of This Work with the Most Closely Related Studies. Key distinctions are highlighted in the deployment and context dimensions.

Ref.	Context	XAI	Web App	Best Acc.	Key Limitation
[11]	Chile (secondary)	SHAP	No	99.7%	No deployed tool; Chilean context only
[12]	Brazil (higher ed.)	SHAP	No	—	No deployment; higher education
[16]	Spain (higher ed.)	No	No	90.8%	No XAI; higher education
[22]	Turkey (secondary)	No	No	94.8%	No XAI or deployment
[17]	Mexico (higher ed.)	No	No	93.4%	No XAI; higher education
[37]	Multiple	SHAP	No	—	No deployment; no Peruvian context
This work	Peru (secondary)	SHAP	Yes	97.27%	Single school; small usability sample

prediction and SHAP explainability; and (3) incorporating 20 locally validated factors including travel time and family structure.

III. EDUPREDICT MODEL

A model integrating ML and XAI is proposed to predict students at risk of school dropout using historical data

(academic performance, socioeconomic factors) and explain the causes of the prediction results. This dual approach allows not only identifying cases with high accuracy but also underpinning pedagogical interventions through interpretive explanations. This model consists of 7 components: data collection, preprocessing, ML model, predictor, XAI model, explainability, and visualization of results. This approach has demonstrated its effectiveness in complex domains such as medicine [38], supporting its potential to address school dropout.

In the model (see Fig. 1), data from students are first collected (Data Collection). Then, preprocessing is performed to clean the data and obtain better results (Preprocessing). Subsequently, the data are processed through a previously trained learning model (ML Model) to obtain the dropout prediction (Predictor). Next, the data and results are processed through a previously trained explainability model (XAI model) to identify the factors that most influenced the prediction (Explainability). Finally, the results are displayed in a clear and easy-to-understand manner for authorities to take corrective measures (Results).

The integration of explainability techniques, such as SHAP, into school dropout prediction models not only allows predicting dropout with accuracy but also understanding the underlying reasons for each prediction. As [12] point out, the ability to correctly interpret the results of a predictive model is crucial and, in some applications, as important as the result itself. On the other hand, according to [11], SHAP allows estimating the contribution of each variable to predictions in a solid, consistent, and highly accurate manner. This feature is essential for educators to implement targeted and effective interventions, based on understanding the underlying causes of dropout.

In Table 5, each of the model components is described.

The Data Collection component incorporates empirically validated risk variables (Table 1), whose association with school dropout is widely documented, ensuring a solid basis for predictive accuracy. For the ML Model component, the algorithms Random Forest, XGBoost, Gradient Boosting, SVM, and KNN were selected, as they have demonstrated superior performance in heterogeneous educational contexts, as evidenced by [8] in Brazil and [9] in virtual environments. Finally, in the Explainability component, the SHAP technique was implemented, due to its ability to quantify the influence of each variable in complex—even non-linear—models, a critical advantage in multifactorial educational systems like the Peruvian one, as supported by [11] and [12].

IV. EDUPREDICT APPLICATION

The EduPredict web application was developed as a comprehensive tool for the prediction and explainability of school dropout in educational institutions. Based on a ML model that, similar to recent studies [39], combines predictive techniques with interpretive methods for actionable results, this system provides school directors and administrators with an accessible and efficient platform to identify and

understand the factors associated with student dropout. Below, the model implementation is detailed, as well as the key functionalities of the application, designed to facilitate informed decision-making in the educational field.

A. ARCHITECTURE

The technological architecture of EduPredict, illustrated in Fig. 2, has been designed to ensure an efficient and robust system in the prediction and explainability of school dropout. The application cohesively integrates front-end and back-end components, which have been deployed in the cloud using Amazon Web Services (AWS) infrastructure. This configuration allows for optimal scalability, high performance, and continuous accessibility for end users, ensuring the delivery of accurate and timely results [40].

TABLE 5. Components of the conceptual model and their functional roles in the EduPredict Framework.

Component	Description
Data Collection	Structured extraction of data on risk factors from institutional academic systems, including grades, attendance, and sociodemographic variables. The quality of these data is critical for detecting predictive patterns of school dropout [11].
Preprocessing	Stage dedicated to optimizing data quality through imputation of missing values, scale normalization, and encoding of categorical variables. Data balancing is also performed on the training partition to prevent leakage. [7] emphasize this step is fundamental to maximizing the effectiveness of ML models. Data are partitioned into training, validation, and test sets.
ML Model	Supervised algorithms recognized for their effectiveness in educational classification tasks are trained using training data and validated with validation data to optimize metrics such as accuracy, precision, and recall.
Predictor	Operational component that applies the trained ML model to estimate dropout probabilities in real time. Its design must ensure high specificity (>90% in cross-validation), guaranteeing that generated alerts correspond to genuine risks [7].
XAI Model	Post-training phase where techniques such as SHAP (SHapley Additive Explanations) or LIME (Local Interpretable Model-agnostic Explanations) are employed to build models that quantify the individual contribution of each factor to the prediction outcomes.
Explainability	Operational component that applies the XAI model to determine the impact of each factor on the dropout prediction for each student. This transparency validates the model's reliability for educators and guides personalized intervention strategies.
Result	This module transforms the model's predictions and explainability analyses into clear visual formats and executive reports, enabling educators to quickly identify at-risk students and understand the primary factors contributing to their dropout prediction.

The architecture diagram illustrates the user's interaction—in this case a school director—with the system through devices such as PCs or laptops, connected to the Internet via technologies like Wi-Fi, Ethernet, or mobile networks (3G, 4G, 5G). This connection infrastructure enables access to the system's core components, starting with the front-end where the web application is deployed on AWS and accessible through a browser, providing an interactive interface for managing school predictions. In the back-end, implemented

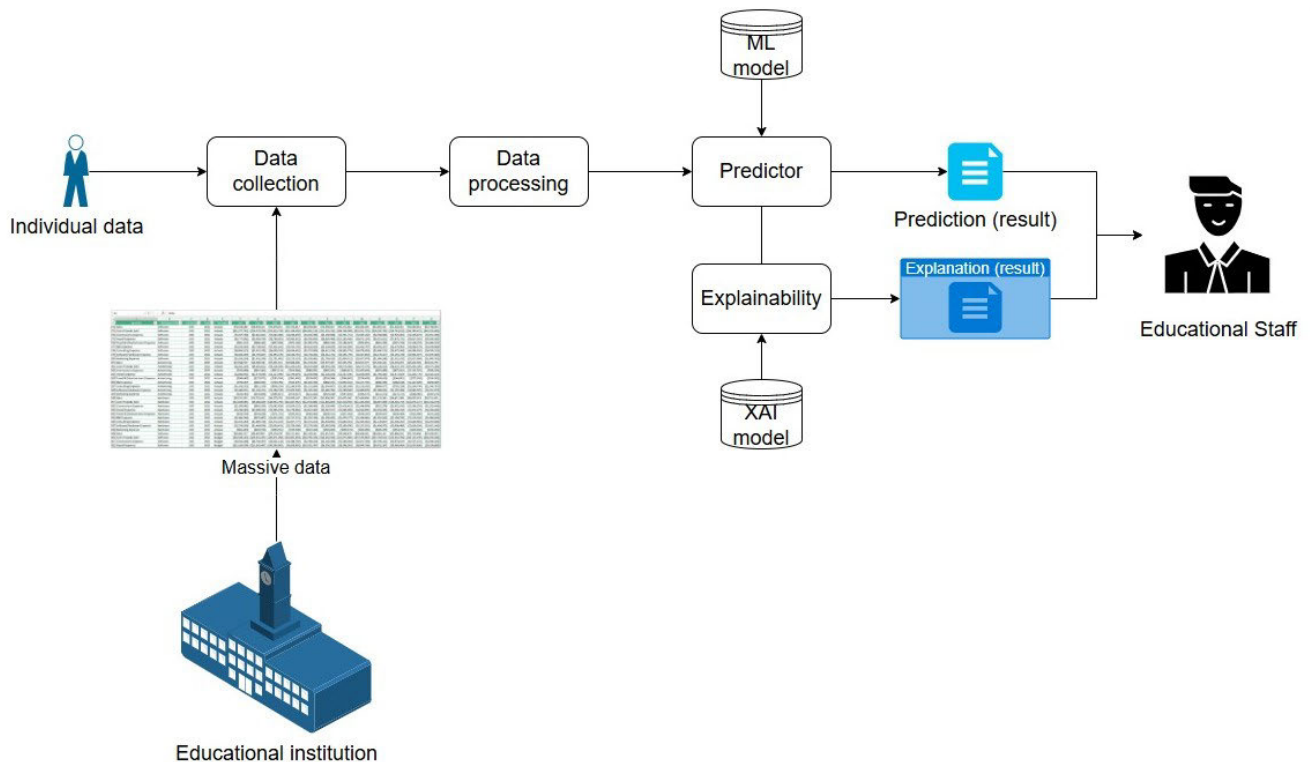


FIGURE 1. Conceptual Model for the Prediction and Explainability of School Dropout. The framework comprises seven sequential components: Data Collection, Preprocessing, ML Model training, Predictor (real-time risk estimation), XAI Model, Explainability (factor contribution analysis), and Results visualization.

in Amazon Beanstalk, the system logic is managed and data stored in a database hosted on Amazon RDS is consumed. This architecture allows the user to perform queries and access prediction results through REST API operations, ensuring connectivity, cloud scalability, and efficient processing of large volumes of data.

B. DATABASE

The application uses a relational database in PostgreSQL, justified by PostgreSQL's ability to optimize complex queries that integrate analytical models, as demonstrated by [41]. This database includes the *users* tables (teachers, directors), which stores personal information and access credentials; *students*, where academic, personal, and risk factor data for each student are stored; *uploaded files*, which records the data files uploaded by users; and *predictions*, which stores the risk predictions generated by the system.

C. DEVELOPMENT

1) FRONT-END

For the front-end development, React.js was used, a widely used JavaScript framework that allows the creation of dynamic and efficient interfaces for web applications [42]. This technology was selected for its flexibility and ability to build reusable components, facilitating both development and maintenance of the application. Additionally, React.js was used to efficiently manage the interface state. The

front-end communicates with the back-end through REST API, ensuring fluid and coherent interaction between the user and the system.

2) BACK-END

The back-end was implemented using AWS, with key services such as Amazon Beanstalk for infrastructure management and Amazon RDS for the database [43]. Amazon Elastic Beanstalk simplifies application deployment and scalability by abstracting infrastructure management, allowing developers to focus on application logic rather than operational overhead. Furthermore, communication between the front-end and back-end is handled through RESTful APIs, ensuring efficient data exchange and reliable bidirectional communication. The use of Elastic Beanstalk as a deployment mechanism is supported by recent studies that highlight its effectiveness in optimizing application performance and reducing deployment latency in cloud-based architectures [44]. Amazon RDS is used to create and manage PostgreSQL database instances and configure, operate, and scale them in the cloud. Additionally, because it offers high availability and automatic scalability, essential features for handling large volumes of data [45], which are essential in educational institutions.

The development was implemented in a unified environment that integrates Visual Studio Code as the main IDE, Git/GitHub for collaborative version control, and pipenv

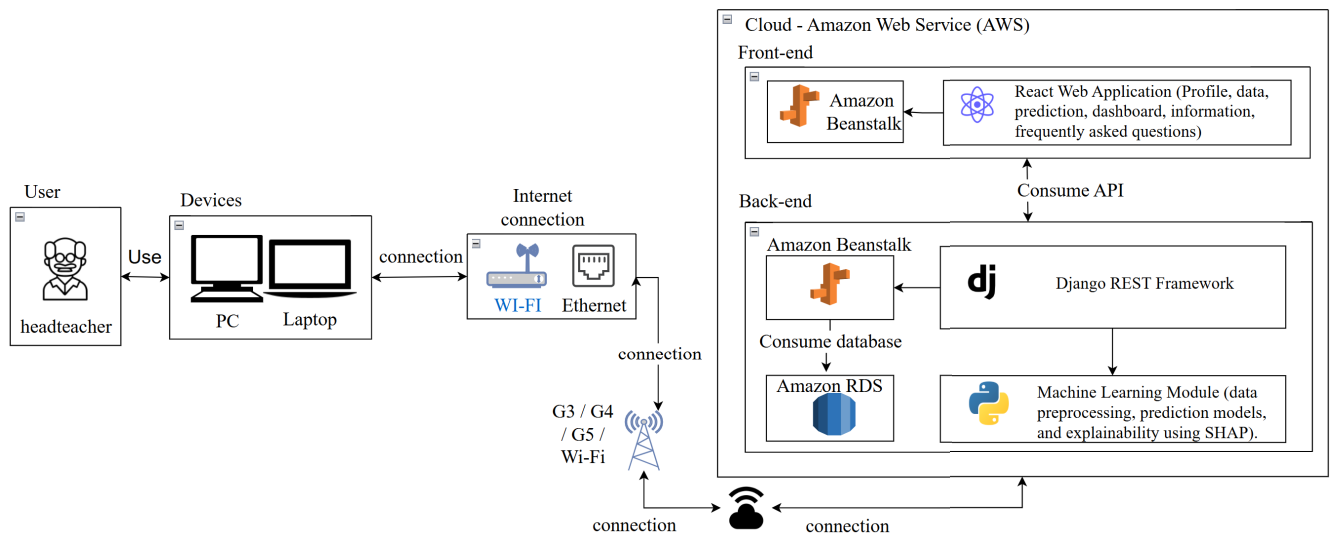


FIGURE 2. Architecture of the EduPredict Application. The system integrates a React.js front-end with a Django back-end deployed on AWS (Elastic Beanstalk and RDS), connected via RESTful APIs.

for managing Python dependencies in an isolated and reproducible manner. The back-end was built with Python and the Django framework, selected for its ability to create scalable and secure RESTful APIs, while the front-end was designed with React.js, facilitating modular and reusable components. For data processing, Pandas was used, and ML models were trained using Scikit-learn (Random Forest, LightGBM), complemented with SHAP for explainability in predictions. Integration with cloud services was done through AWS SDK, connecting the application with RDS and Beanstalk to ensure scalability and availability.

D. MODULES

The application consists of four modules: Data Collection, Predictor, Explainability, Information, Frequently Asked Questions, and User Management. Below, the first three modules are described, where the Predictor and Explainability modules implement the proposed conceptual model. The results mentioned in the conceptual model are included in the Predictor and Explainability modules.

1) DATA COLLECTION MODULE

This module allows the user to upload student information from an Excel file. These data include academic, psychosocial, socioeconomic, and sociodemographic factors, which are used by the machine learning model to generate predictions. The system automatically validates the uploaded data, ensuring they meet format requirements, with no duplicates or errors that could compromise the accuracy of the results. If the file does not meet the requirements, the system will display a message specifying the non-compliance, for example, “invalid format. Only Excel files (xlsx, xls) are accepted.” This process simplifies the management of large

volumes of information and significantly reduces the margin of error in data entry.

2) PREDICTOR MODULE

This module constitutes the core of the platform, where data are first preprocessed automatically after they have been uploaded and validated, and immediately the machine learning model is applied to predict the risk of school dropout. The system uses advanced algorithms to analyze the provided data and generate accurate predictions, classifying students into categories of “at risk” or “not at risk.” The results are presented in an intuitive interface that allows users to view individual and group predictions, facilitating the identification of patterns and trends in the student population.

Fig. 4 illustrates the interface of the predictor module, where the results of the dropout risk analysis generated by the ML model are visualized. In the upper section, the option to delete previous predictions (DELETE PREDICTION) and filters to search for students by name or DNI are shown. The central table presents the list of students with their respective risk percentages and assigned categories (High Risk, Moderate, Low). In the upper right part, the system allows PERFORM PREDICTION for new analyses and DOWNLOAD REPORT in Excel with the consolidated results, facilitating the prioritization of interventions.

3) EXPLAINABILITY MODULE

The explainability module employs techniques based on SHAP to provide detailed interpretations of the predictions generated by the machine learning model, which is reported in a PDF file. This module is fundamental for users to understand the underlying reasons for each prediction, allowing them to make informed and evidence-based decisions. The system breaks down the model’s predictions into individual

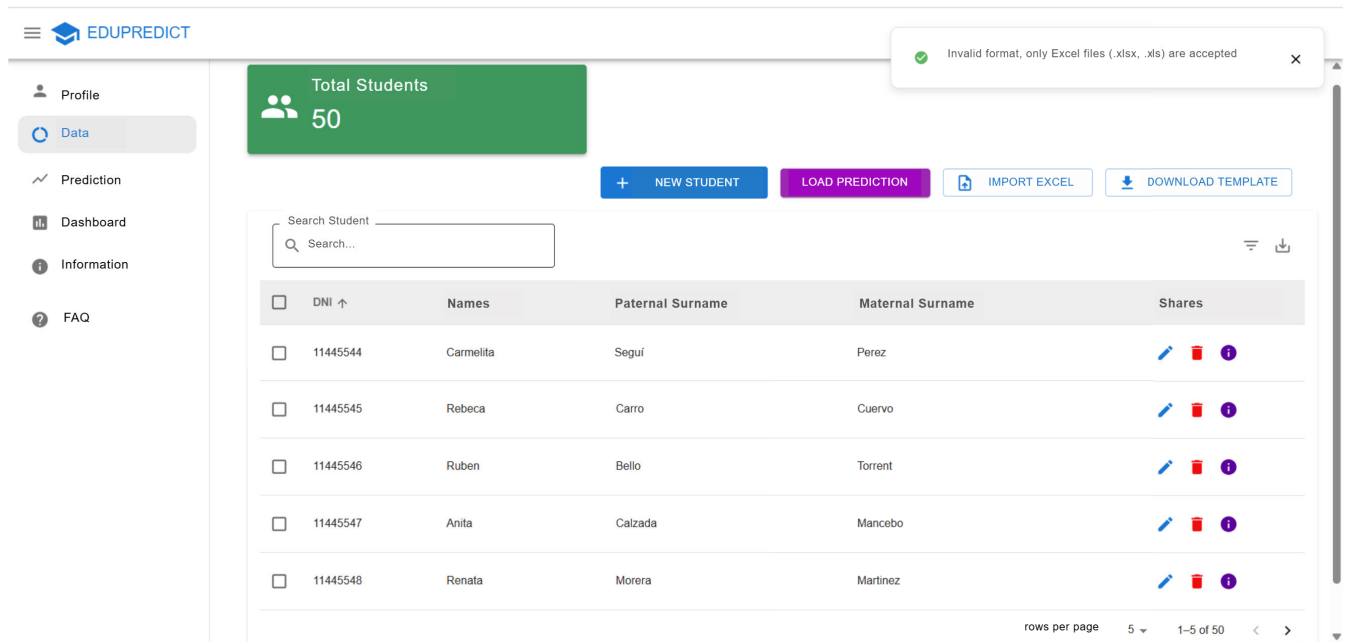


FIGURE 3. Data collection module interface. The module allows users to upload student records from Excel files containing academic, psychosocial, socioeconomic, and sociodemographic variables used as inputs for the machine learning dropout prediction model.

contributions from each variable, offering a clear and intuitive visualization of the factors that most influence the dropout risk.

Fig. 5 shows the results generated by the explainability module using SHAP, highlighting the most influential features in the dropout risk at a general level. In the upper part, a quantitative summary of students categorized by risk level (36 High, 7 Moderate, 7 Low, out of a total of 50) is presented. The central section, titled “Most Influential Features (SHAP),” uses average SHAP value graphs to visualize the relative weight of each variable in the model’s predictions; it also allows selecting a specific student and viewing the factors that affect that student in particular. These graphs enable identifying global patterns, such as which academic or socioeconomic factors have the greatest impact on dropout. The interface facilitates immediate interpretation of the results, fulfilling the objective of transparency.

V. VALIDATION

The proposed model and consequently its implementation is validated through numerical experiments using five ML models to predict the risk of school dropout in students from a school in Metropolitan Lima and the accuracy and efficiency of the models is evaluated. In addition, the usability, utility and satisfaction of the application is measured.

A. DATASET

The dataset used comes from a public secondary school in Metropolitan Lima and consists of 1,464 records with a total of 21 fields of which 20 are factors (see Table 6), each representing key characteristics of the students, and the

remaining field is the student code. The analyzed school is located in a peripheral area of Metropolitan Lima, situated in the urban cones characterized by high population density and social vulnerability conditions. More than 65% of the students reside in the vicinity of the school, traveling on foot or by local transport in an approximate maximum time of 15 minutes. This highlights the community and social integration role that the institution plays within its immediate environment.

These factors include academic, psychosocial, socio-economic and sociodemographic information. Among these factors are the educational levels of the parents, the age and gender of the student. The importance of these factors has been highlighted in various studies. [46] underscore that lower educational levels of parents correlate with higher dropout rates, while [47] and [48] identify gender as a critical indicator when evaluating the risk of disengagement and school dropout during educational transitions.

The data were preprocessed through the following sequential steps: (1) removal of 8 records with missing values, reducing the working dataset from 1,464 to 1,456 records; (2) encoding of categorical factors using `LabelEncoder`; and (3) min-max normalization of all numeric attributes. The original dataset exhibited class imbalance, with 492 dropout cases (33.6%) and 972 non-dropout cases (66.4%), yielding an imbalance ratio of approximately 1:1.97. To address this imbalance, SMOTE (Synthetic Minority Over-sampling Technique) was applied *exclusively to the training partition* to prevent information leakage into the validation set. The test set preserved the original class distribution. After this processing, the working dataset consisted of 1,456 records

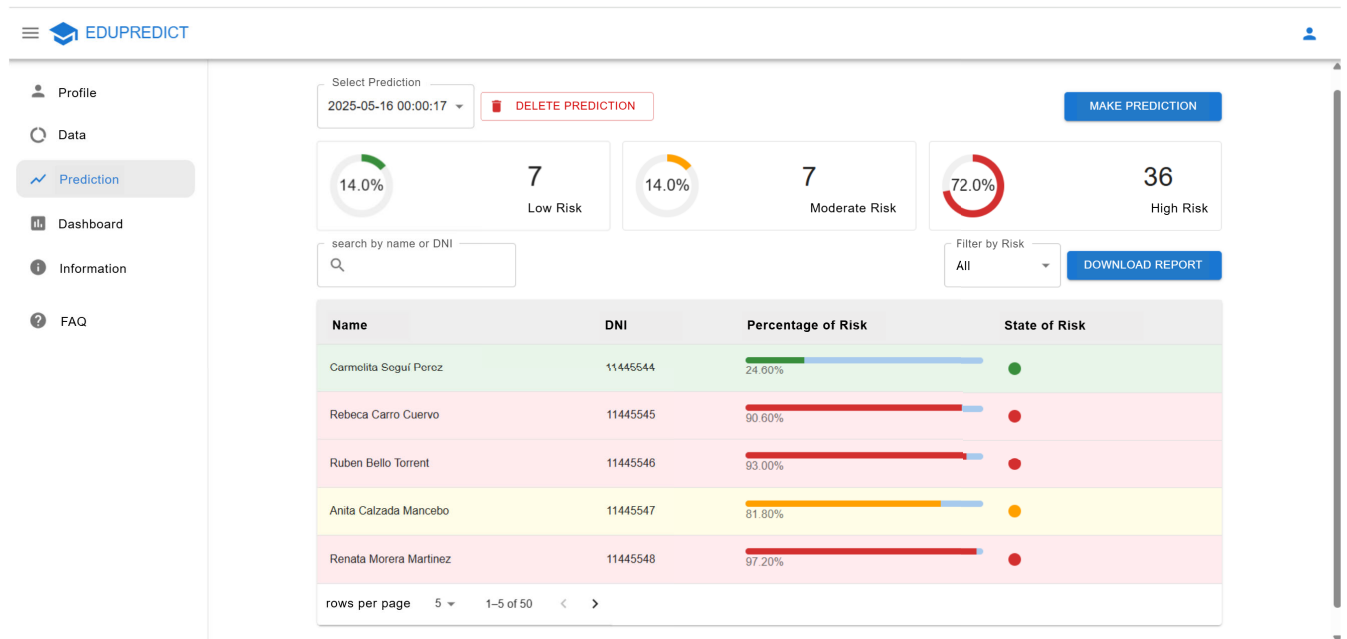


FIGURE 4. Predictor module interface. The system processes validated student data and applies the machine learning model to estimate dropout risk, displaying predicted risk percentages and categories (High, Moderate, Low) for each student.

(see Table 7). Of these, 80% (1,165 records) were allocated for model training (with SMOTE applied to balance classes within the training set) and 20% (291 records) for validation, corresponding to 146 dropout cases and 145 non-dropout cases, allowing effective evaluation of performance under realistic class conditions.

B. PARTICIPANTS

The participants in the system evaluation are 10 professionals ($n = 10$) with experience between 2 and 11 years, of which two hold management positions, two are psychologists, and six are teachers, all from the same public secondary school. The description of the participants is shown in Table 8.

C. INSTRUMENTS

To evaluate the user experience with the application, a usability questionnaire was applied. According to [49], the Likert scale is an effective tool for measuring people's opinions and perceptions on a topic. Therefore, this questionnaire included a series of questions in which participants rated their experience using a Likert scale from 1 to 5, where 1 represents "None", 2 "Little", 3 "Medium", 4 "High", and 5 "Very high". The questions used in the usability questionnaire are detailed in Table 9.

On the other hand, to evaluate the results of the efficiency of the proposed system, the metrics of accuracy, sensitivity, precision, and specificity were used. All these metrics are derived from the confusion matrix, which includes the values of True Positives (TP), True Negatives (TN), False Negatives (FN), and False Positives (FP). The metrics used to evaluate the models are summarized in Table 10.

D. EXPERIMENTS

For the validation of our experiments, five machine learning algorithms implemented with the scikit-learn library in Python were used. Each of these algorithms was optimized through hyperparameter selection using Grid Search, which are detailed in Table 11.

Additionally, 2 scenarios have been considered to know the impact of the factors on the results for the five implemented ML models:

- **First scenario:** The 20 factors from Table 6 were used.
- **Second scenario:** The 5 most influential factors (health, age, family relationship, annual final grade, and behavior) identified through the GreedyStepwise algorithm were used. GreedyStepwise was selected because of its computational efficiency relative to exhaustive search methods: it iteratively adds or removes one feature at a time, evaluating each step against a cross-validated accuracy criterion, thereby identifying high-impact feature subsets without exponential computational cost [51]. This reduced scenario is particularly relevant for contexts with limited data availability, common in Peruvian public schools.

With the objective of evaluating the software usability, a digital survey (Google Forms) was administered to the participants, after a structured process that includes: (1) access registration to the system; (2) a 30-minute practical training session based on demonstrative cases; and (3) a seven-day autonomous use period. Although the tool has an intuitive design, the training was implemented to ensure proper use of the application.

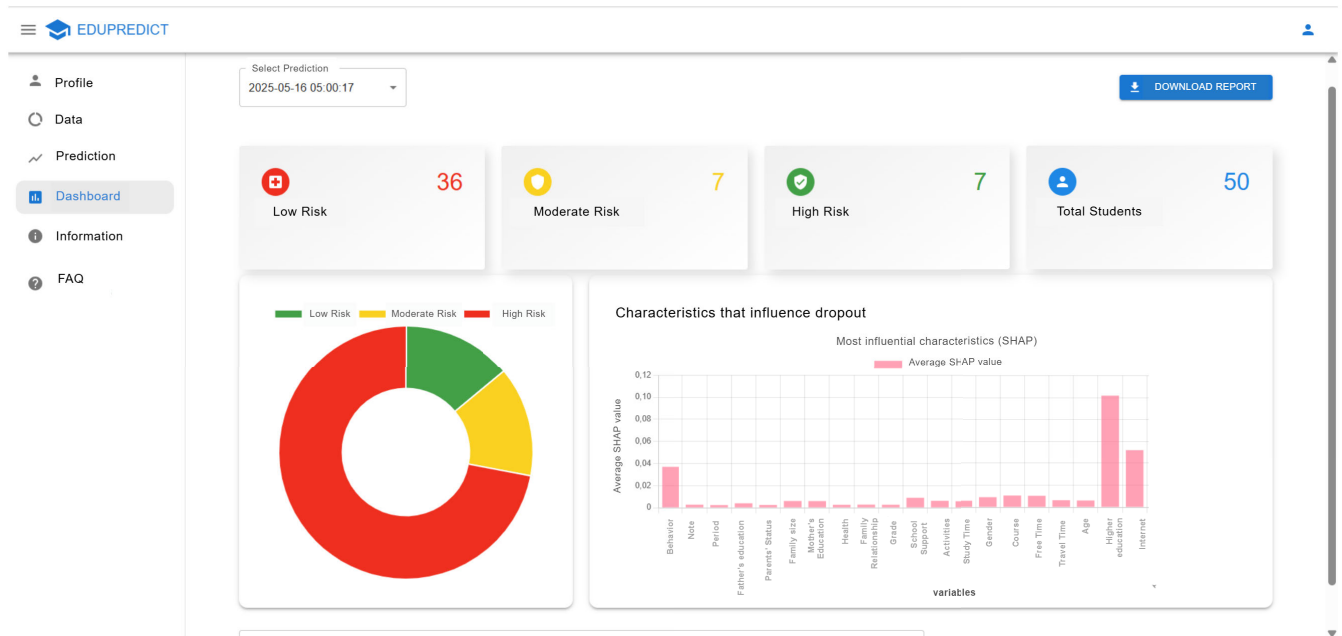


FIGURE 5. Explainability module interface based on SHAP. The module provides global explanations of the machine learning predictions, highlighting the most influential features contributing to students' dropout risk.

E. RESULTS

1) CONFUSION MATRIX

Fig. 6 presents the confusion matrices of the 5 implemented models on the test set (291 records). The matrices display raw (unnormalized) predicted counts organized as: True Positives (dropout correctly predicted), False Negatives (dropout missed), False Positives (non-dropout misclassified), and True Negatives (non-dropout correctly predicted). The color scale represents the absolute number of observations in each cell, where darker shades indicate higher counts and lighter shades indicate lower frequencies. All five models show strong performance, with Gradient Boosting (Fig. 6e) achieving the lowest rate of false negatives (undetected dropout cases), which is critical for timely educational interventions.

2) LOSS FUNCTION

Fig. 7 shows the loss function in the training and validation phase of the 5 implemented models. It is observed that KNN (Fig. 7d) converges faster (around iteration 30); however, its loss function remains above that of Gradient Boosting (GBC) (Fig. 7e). Importantly, all models show convergent training and validation loss curves without a notable divergence, indicating that overfitting was not a significant concern under the applied regularization and cross-validation strategy.

3) PERFORMANCE

Table 12 presents the performance of the 5 implemented models for the two considered scenarios. Gradient Boosting (GBC) stands out as the best model, achieving 99.28% precision and 99.66% specificity in the first scenario, slightly

surpassing XGBoost (98.57% precision). This result is due to the intrinsic capacity of boosting methods (like GBC and XGBoost) to handle non-linear relationships between variables through the iterative combination of weak trees, minimizing the residual error in each iteration—particularly useful with complex data (20 factors). Additionally, its focus on optimizing loss functions explains its superiority over models like SVM or KNN. In the reduced scenario of 5 factors, Random Forest and XGBoost stand out with accuracy greater than 95%, and specificity greater than 96%. The results confirm that boosting-based models (GBC, XGB) are superior for predicting dropout with complex data, while RF performs well with reduced data. Balanced accuracy (BA), computed as the arithmetic mean of sensitivity and specificity, is also reported. It is important to note that BA is a threshold-dependent metric that provides a single-point approximation and is not equivalent to the full area under the ROC curve (AUC), which integrates performance across all decision thresholds [50].

4) EXPLAINABILITY

a: CORRECT PREDICTION OF DROPOUT (D1)

A student who dropped out and whom the model correctly predicted was analyzed. He was taking the second academic period (F01: Period = 1), was in a low school grade (F02: Grade = 0.25), was male (F04: Gender = 1), belonged to a large family (F05: Family Size = 1), and his parents do not live together (F06: Parents' Status = 0). The father had a higher educational level (F07: Father's Education = 1), while the mother had secondary studies (F08: Mother's Education = 1). He had a travel time within the established

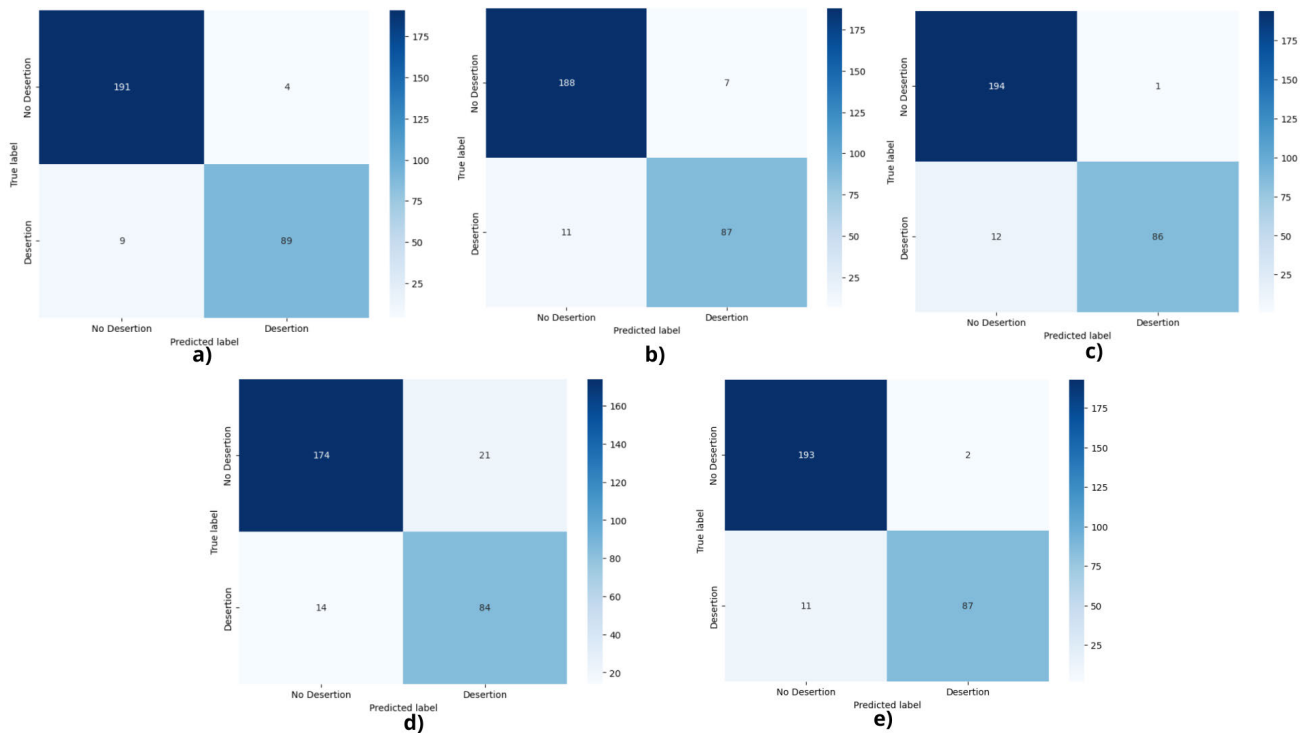


FIGURE 6. Confusion matrices (raw counts, test set) for the first scenario (20 factors). Each matrix displays true negatives (top-left), false positives (top-right), false negatives (bottom-left), and true positives (bottom-right). Darker shades represent higher counts. Models: a) XGBoost; b) SVM; c) Random Forest; d) KNN; e) Gradient Boosting.

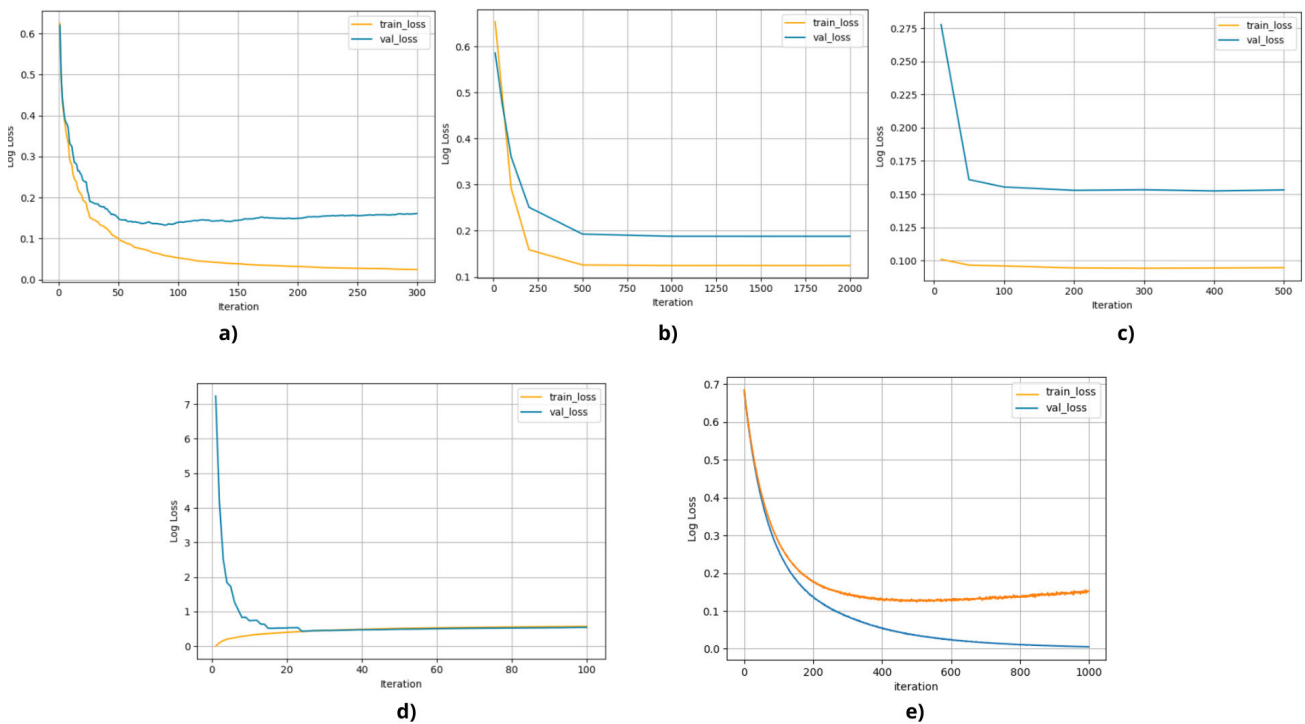


FIGURE 7. Training and validation loss curves across iterations for the five machine learning models. The orange line corresponds to training loss and the blue line to validation loss. The generally convergent trajectories, without pronounced divergence between the curves, suggest the absence of severe overfitting. Models: a) XGBoost; b) SVM; c) Random Forest; d) KNN; e) Gradient Boosting.

ranges (F09: Travel Time = 2) and little study time (F10: Study Time = 1). His school support was limited

(F11: School Support = 1), as was his participation in extracurricular activities (F12: Activities = 1). He had

TABLE 6. Risk factors and types of selected attributes used as input to the ML Models.

ID	Factor	Type
F01	Period	Numeric
F02	Grade	Numeric
F03	Age	Numeric
F04	Gender	Categorical
F05	Family Size	Categorical
F06	Parents' Status	Categorical
F07	Father's Education	Categorical
F08	Mother's Education	Categorical
F09	Travel Time	Categorical
F10	Study Time	Categorical
F11	School Support	Categorical
F12	Extracurricular Activities	Categorical
F13	Higher Education Aspiration	Categorical
F14	Internet Access	Categorical
F15	Family Relationship	Categorical
F16	Free Time	Categorical
F17	Health Status	Categorical
F18	Behavior	Numeric
F19	Course	Categorical
F20	Grade	Numeric

F01: 0 (First trimester), 1 (Second), 2 (Third), 3 (Fourth); F02: 0, 0.25, 0.5, 0.75, 1; F03: 0.73 (12y), 0.8 (13y), 0.87 (14y), 0.93 (15y), 1 (16y), 0 (17y)—the value 0 for 17-year-old students is intentional: in the Peruvian secondary system the normative age range is 12–16 years; students aged 17 are over-age, a condition associated with elevated dropout risk, and are therefore encoded as the highest-risk category; F04: 0 (female), 1 (male); F05: 0 (>3), 1 (≤3); F06: 0 (Separated), 1 (Together); F07–F08: 0 (Primary), 1 (Secondary), 2 (Higher), 3 (None); F09: 0 (15–30 min), 1 (30–60 min), 2 (>1 hr), 3 (≤15 min); F10: 0 (2–5 hrs), 1 (<2 hrs); F11–F12: 0 (No), 1 (Yes); F13–F14: 0 (Yes), 1 (No); F15: 0 (Poor), 1 (Excellent), 2 (Very poor), 3 (Good), 4 (Fair); F16: 0 (2–5 hrs), 1 (>5 hrs), 2 (≤2 hrs); F17: 0 (Poor), 1 (Good), 2 (Very poor), 3 (Excellent), 4 (Fair); F18: 0 (18–20), 1 (10–11), 0.75 (12–13), 0.5 (14–15), 0.25 (16–17); F19: 0 (Mathematical Reasoning), 1 (History of Peru), 2 (Verbal Reasoning), 3 (English); F20: 0 (17–20), 1 (5–10), 0.67 (11–13), 0.33 (14–16).

TABLE 7. Characteristics of the Original and Preprocessed Dataset. The balanced counts in the preprocessed column reflect the training partition after SMOTE; the test set retained the original distribution.

Original Dataset	Preprocessed Dataset
Number of records = 1,464	Number of records = 1,456
Number of factors = 20	Number of factors = 20
Data type = Categorical and Numeric	Data type = Numeric
Dropout students = 492 (33.6%)	Dropout students = 728 (training, after SMOTE)
Non-dropout students = 972 (66.4%)	Non-dropout students = 728 (training, after SMOTE)

TABLE 8. Profile of the $n = 10$ Participants Who Evaluated the EduPredict application.

ID	Age	Gender	Exp. (yr)	Degree	Position
U01	49	Male	11	Bachelor's	Principal
U02	55	Male	9	Bachelor's	Vice Principal
U03	38	Male	2	Bachelor's	Psychologist
U04	46	Female	4	Bachelor's	Psychologist
U05	35	Male	3	Bachelor's	Teacher
U06	30	Male	2	Bachelor's	Teacher
U07	41	Male	5	Bachelor's	Teacher
U08	45	Female	4	Bachelor's	Teacher
U09	33	Male	2	Bachelor's	Teacher
U10	37	Male	5	Bachelor's	Teacher

a family member with higher education (F13: Higher Education = 1) and access to internet (F14: Internet = 1). He

TABLE 9. Usability Form Questions Organized by Dimension (Usability, Utility, Satisfaction).

Dimension	ID	Question
Usability	Q1	The application's interface is intuitive.
	Q2	The application's functions are easy to find.
	Q3	The application provides me with relevant information about school dropout.
	Q4	The explainability of the results is clear and understandable.
Utility	Q5	The predictions made by the system are useful for making educational decisions.
	Q6	The application helps identify students at risk of dropout.
Satisfaction	Q7	My overall experience with the application has been positive.
	Q8	I would like to continue using this application in the future.
	Q9	I am satisfied with the functionalities offered by the application.
	Q10	I have encountered technical errors or malfunctions.

TABLE 10. Metrics Used to Evaluate the ML Models, Derived from the confusion matrix.

Metric	Description	Formula
Accuracy	Rate of students whose school dropout was correctly predicted, compared to the total.	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	Rate of students who actually dropped out and were correctly classified as dropouts.	$\frac{TP}{TP+FP}$
Sensitivity	Rate of true dropouts who were correctly identified.	$\frac{TP}{TP+FN}$
Specificity	Rate of students who did not drop out and were correctly classified as non-dropouts.	$\frac{TN}{TN+FP}$
Bal. Accuracy	Arithmetic mean of sensitivity and specificity. Note: BA is a threshold-dependent metric and is not equivalent to the full AUC statistic computed across all thresholds; it serves only as a single-point approximation [50].	$\frac{Sensitivity+Specificity}{2}$

TABLE 11. Hyperparameters of the ML Models selected via grid search cross-validation.

Model	Final Hyperparameters
Random Forest	bootstrap = True, max_depth = 10, min_samples_split = 2, min_samples_leaf = 1, n_estimators = 200, class_weight = balanced, max_features = sqrt
XGBoost	n_estimators = 700, learning_rate = 0.1, max_depth = 5, colsample_bytree = 0.6, subsample = 0.6, gamma = 0, min_child_weight = 1
SVM	C = 1.0, kernel = rbf, gamma = scale, degree = 3
Gradient Boosting	n_estimators = 500, learning_rate = 0.1, max_depth = 3, min_samples_split = 2, min_samples_leaf = 1, subsample = 0.6, max_features = sqrt
KNN	n_neighbors = 5, weights = distance, metric = euclidean, p = 2

presented a deteriorated family relationship (F15: Family Relationship = 1) and reported little free time (F16: Free Time = 0). Additionally, he presented non-optimal health conditions (F17: Health = 1), problematic behavior (F18:

TABLE 12. Results Obtained after evaluating the models in two scenarios. BA (balanced accuracy) is a threshold-dependent metric, not equivalent to the full AUC.

Model	Scenario 1 (20 factors)					Scenario 2 (5 factors)				
	Acc	Pre	Sen	Spe	BA	Acc	Pre	Sen	Spe	BA
RF	95.91	97.12	90.60	98.63	94.62	95.23	96.38	89.26	98.28	93.77
SVM	92.27	88.08	89.26	93.81	91.54	94.55	90.32	93.96	94.85	94.41
GBC	97.27	99.28	92.62	99.66	96.14	94.55	93.10	90.60	96.56	93.58
KNN	93.81	91.12	98.96	93.92	96.44	94.77	95.00	89.26	97.59	93.43
XGB	97.65	98.57	92.62	99.31	95.97	95.00	93.79	91.28	96.91	94.10

RF: Random Forest; SVM: Support Vector Machine; GBC: Gradient Boosting Classifier; KNN: K-Nearest Neighbors; XGB: XGBoost. Acc: Accuracy (%); Pre: Precision (%); Sen: Sensitivity (%); Spe: Specificity (%); BA: Balanced Accuracy (%) = (Sen+Spe)/2.

TABLE 13. Information on students' Risk factors in three different scenarios (D1: Correct Dropout Prediction; D2: Correct Non-Dropout Prediction; D3: Incorrect dropout prediction).

Factor	D1. Correct dropout	D2. Correct non-dropout	D3. Incorrect dropout
F01	1	1	1
F02	0.25	0.25	0.25
F03	0.8	0.8	0.8
F04	1	0	0
F05	1	1	1
F06	0	1	0
F07	1	1	1
F08	1	1	1
F09	2	0	0
F10	1	1	0
F11	1	0	1
F12	1	0	0
F13	1	1	0
F14	1	1	1
F15	1	0	1
F16	0	0	0
F17	1	0	4
F18	1	0.25	0.5
F19	2	2	1
F20	1	0	0.33

Behavior = 1) and was taking a certain subject (F19: Course = 2). Finally, he showed low academic performance (F20: Grade = 1). This information on risk factors for this and two additional scenarios is presented in Table 13.

With that data, the SHAP model obtained the results presented in Fig. 8, where the size of the red bars indicates the level of positive influence of a factor on dropout and the blue bars the negative influence on no dropout. Table 14 provides a numerical summary of the top SHAP contributions for all three scenarios.

According to SHAP, the factors that contributed the most to dropout were behavior (F18: +15.44) and grade (F20: +14.40), followed by academic period (F01: +1.04), family size (F05: +0.75), mother's education (F08: +0.71), health (F17: +0.68), family relationship (F15: +0.61), school grade (F02: +0.53), and study time (F10: +0.22). Other factors with slight positive impact were travel time (F09: +0.10), age (F03: +0.10), and internet (F14: +0.00). In the opposite direction, some variables reduced the estimated probability of dropout, such as school support (F11: −0.35), extracurricular activities (F12: −0.32), gender (F04: −0.14), course (F19: −0.12), free time (F16: −0.11), higher education (F13:

TABLE 14. Numerical SHAP Contributions (Top Factors) for the Three Analyzed Scenarios. Positive values indicate contribution toward dropout; negative values indicate contribution toward non-dropout.

Factor	ID	D1 (SHAP)	D2 (SHAP)	D3 (SHAP)
Behavior	F18	+15.44	−9.85	−0.72
Grade (annual)	F20	+14.40	−6.32	−5.22
Father's Education	F07	−0.90	+1.79	+0.51
Parents' Status	F06	−0.78	+0.43	−0.29
Family Relationship	F15	+0.61	−0.31	+2.52
Health Status	F17	+0.68	−0.12	−1.69
Higher Ed. Aspiration	F13	−0.09	+0.72	+1.88
School Support	F11	−0.35	−0.24	−1.37
Mother's Education	F08	+0.71	+0.72	+0.44
Academic Period	F01	+1.04	−0.18	−0.11
Cumul. toward dropout		92.32%	20.48%	48.95%
Cumul. toward no dropout		7.68%	79.52%	51.05%

−0.09), parents' marital status (F06: −0.78) and, with greater negative effect, father's education (F07: −0.90). Overall, the model identifies that the combination of academic, behavioral, and family factors has a decisive weight in the risk of school dropout.

b: CORRECT PREDICTION OF NO DROPOUT (D2)

A student who did not drop out and whose permanence was correctly anticipated by the predictive model was analyzed. This student presents information on risk factors described in Table 13. With this data, the SHAP model generated the results observed in Fig. 9, where the factors that most influenced the student's permanence (no dropout) are F18 (behavior) and F20 (grade) with influence of −6.32 and −9.85 respectively, while in lesser proportion the factors that most influenced dropout are F07 (mother's education) and F19 (course) with degree of influence of +1.79 and +0.72 respectively. The figure also shows a greater degree of influence toward permanence (no dropout) than dropout, with accumulated values of 19.69 and 5.07 respectively, which confirms coherence between the prediction, the real result, and the explanation.

c: INCORRECT PREDICTION OF DROPOUT (D3)

The case of a student whose data are presented in Table 13, who did not drop out but whom the model erroneously classified as a dropout, was analyzed. Fig. 10 shows that the factors that most influenced the student's permanence (no dropout) are F17 (Health) and F20 (Grade), with influences of −1.69 and −5.22 respectively, while the factors that most drove the classification toward dropout were F13 (Higher Education) and F15 (Family Relationship), with degrees of influence of +1.88 and +2.52 respectively. The accumulated degree of influence of factors associated with permanence reaches 11.91, while that of factors related to dropout is 11.42. This near-balance between the two opposing sets of influences explains the erroneous classification: the minimal difference between accumulated SHAP scores caused the model to predict dropout despite the student's actual continuation.

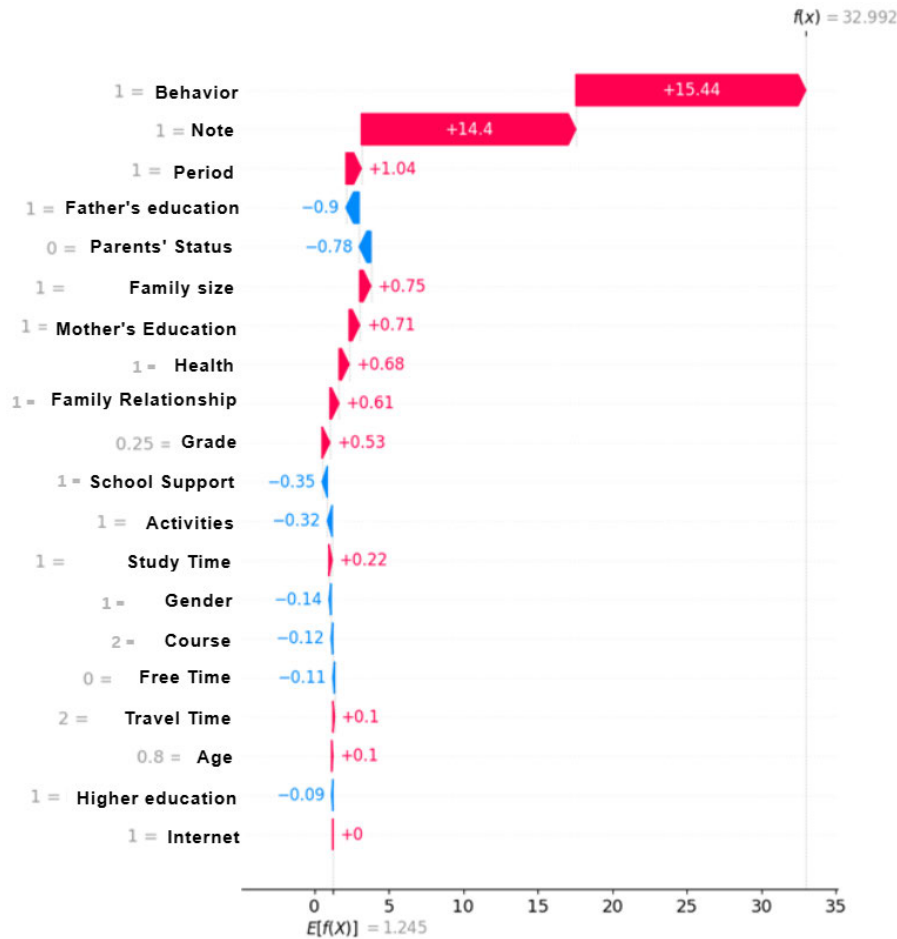


FIGURE 8. SHAP waterfall plot for the correct dropout prediction scenario (D1). Red bars indicate factors pushing the prediction toward dropout; blue bars indicate factors pushing toward non-dropout. Bar length represents the magnitude of each factor's SHAP contribution.

5) USABILITY, UTILITY AND SATISFACTION

Table 15 shows the results of the usability, utility and satisfaction survey of the EduPredict application applied to the $n = 10$ participants, reporting the mean and standard deviation (SD) per question and the mean per analysis dimension.

VI. DISCUSSION

In general, the evaluated ML models (RF, GBC, KNN, and XGB) using a dataset of 1,456 records from a Peruvian public school (80% training with SMOTE, 20% validation with original distribution) exhibited superior performance in the 20-factor scenario compared to the 5-factor scenario. However, this improvement was marginal, with average differences less than 0.6% in accuracy and specificity, and less than 1.95% in precision and sensitivity. In contrast to this trend, SVM showed better performance in the reduced 5-factor scenario, behavior attributable to the implemented RBF kernel, which is sensitive to redundant or correlated variables [51]. This limitation is mitigated through

dimensionality reduction with the GreedyStepwise method, which eliminates noise and redundancies, allowing SVM to improve pattern identification. Additionally, the results confirm the marginal superiority of boosting-based models (GBC, XGB), due to their ability to handle non-linear relationships through the iterative combination of weak trees, minimizing residual error.

In the five-factor scenario, the RF and KNN models present strong results with accuracy, precision, and specificity above 94.5%, 95.0%, and 97.5% respectively. This five-factor scenario (health, age, family relationship, annual final grade, and behavior) is appropriate when there is a scarcity of data availability, as occurs in many public schools in developing countries like Peru.

A. ADDRESSING THE HIGH ACCURACY VALUES

The reported accuracy values (95–97%) may appear elevated; however, several design decisions mitigate overfitting risk. First, SMOTE was applied exclusively to the training partition; the test set preserved the original class distribution,

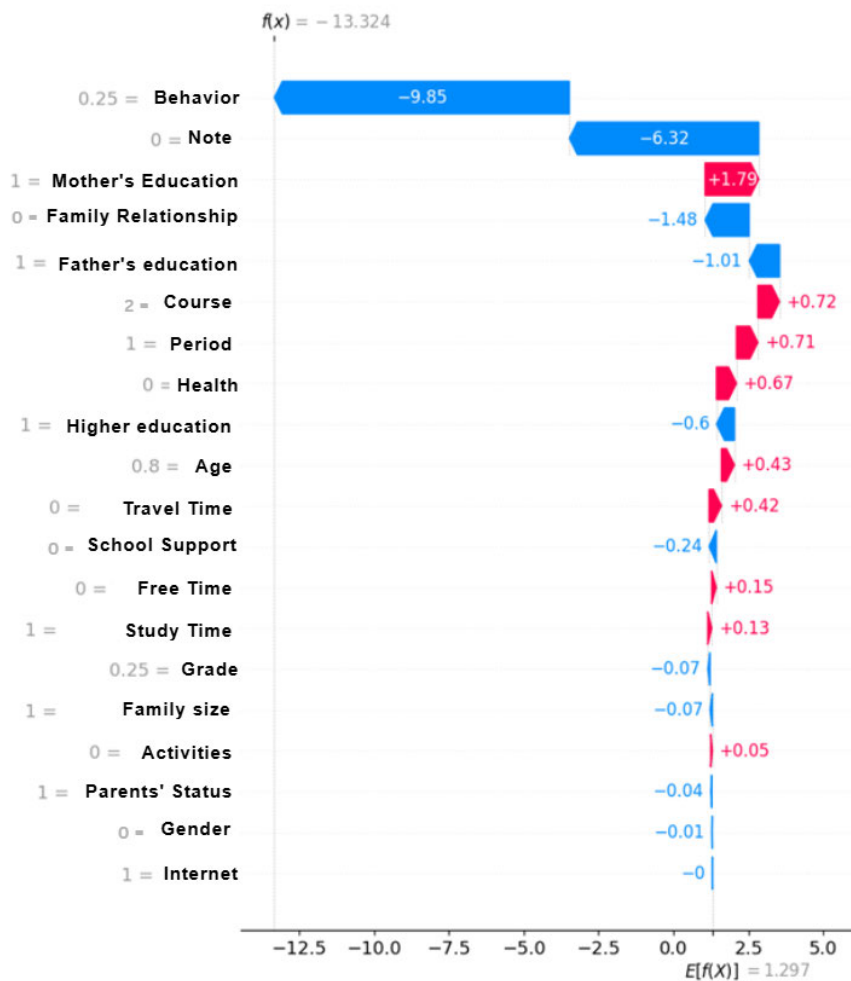


FIGURE 9. SHAP waterfall plot for the correct non-dropout prediction scenario (D2). Blue bars (left) indicate protective factors contributing to the prediction of non-dropout; red bars (right) indicate risk factors.

providing an unbiased performance estimate. Second, all hyperparameters were selected via Grid Search cross-validation on the training set only, without exposure to test data. Third, the loss function curves (Fig. 7) show convergent training and validation trajectories for all models, with no visible divergence between the two curves—a standard indicator of the absence of severe overfitting. Fourth, comparably high performance values have been reported for structured, well-defined educational datasets in similar studies [11], [17], where the small number of informative features and the structured nature of the data can naturally yield high separability between classes. Nonetheless, we acknowledge that the high precision values (especially GBC at 99.28%) should be interpreted cautiously in the context of a single-school dataset; validation on broader, multi-school datasets remains essential.

To address the balance between predictive accuracy and interpretability, the SHAP method was used to improve transparency and understand school dropout predictions. This

approach offers local explanations specific to each student, facilitating its use by educators. The applicability of SHAP was evaluated in three scenarios through case studies:

- **Correct Prediction of Dropout:** The explanation was robust, highlighting eleven risk factors, such as behavior (+15.44) and grade (+14.40), which contributed 92.32% to the dropout risk, compared to eight factors like father's education (−0.90) and school support (−0.35), which added 7.68% of contrary influence.
- **Correct Prediction of No Dropout:** The explanation was equally solid, with eleven protective factors, such as behavior (−9.85) and grade (−6.32), representing 79.52% of the influence toward permanence, compared to nine risk factors, such as mother's education (+1.79) and course (+0.72), which contributed 20.48%.
- **Incorrect Prediction of Dropout:** Although infrequent due to the model's accuracy, this case showed that seven protective factors, such as academic performance (−5.22) and school support (−1.37), contributed

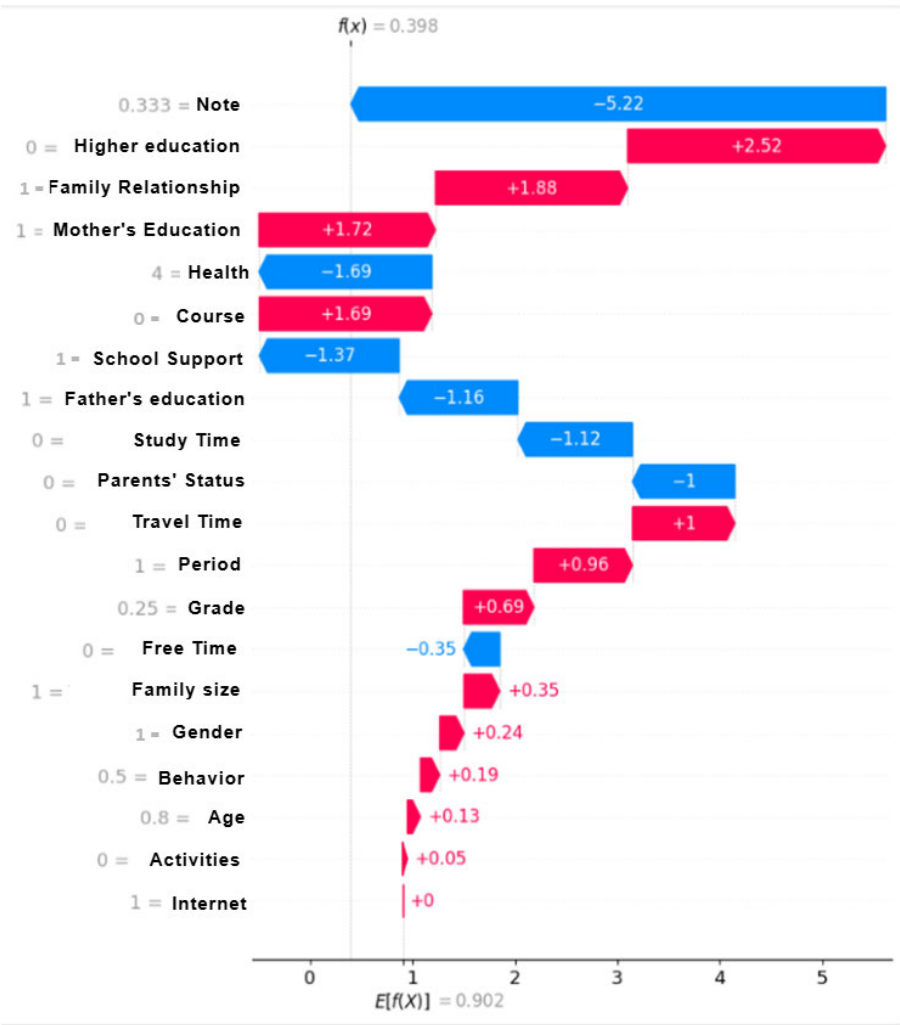


FIGURE 10. SHAP waterfall plot for the incorrect dropout prediction scenario (D3). The near-balance between protective factors and risk factors explains the misclassification.

51.05% to reducing the risk, compared to twelve risk factors, such as higher education of a family member (+2.52) and family relationship (+1.88), which added 48.95%.

The consistency of SHAP explanations is validated in the three scenarios. Following the approach given in the study by [52], it was observed that, in correct predictions, the probability ratio exceeded 1, with values of 11.02 $[(92.32 - 7.68)/7.68]$ for dropout and 2.88 $[(79.52 - 20.48)/20.48]$ for no dropout. In the incorrect prediction, the ratio was 0.04 $[(51.05 - 48.95)/48.95]$, indicating high imprecision due to the balance between factors. These results support the model's robustness and its utility for designing personalized educational interventions, such as academic reinforcement or family mediation, based on the identified factors.

EduPredict was implemented with the objective of providing a simple and accessible tool for the prediction and explainability of school dropout, integrating predictive models (RF, SVM, GBC, KNN, and XGB) and an

explainability model (SHAP). The technological architecture is based on React for the frontend and Django in the backend, using a PostgreSQL database. The system was deployed on Amazon Web Services (AWS) cloud, leveraging the EC2 service for infrastructure and Amazon RDS for database management. The application allows uploading Excel documents and viewing results in PDF and Excel formats, providing educators with practical tools for informed decision-making.

B. PRACTICAL IMPLICATIONS

The EduPredict framework offers concrete value for educational practice at multiple levels. At the student level, the per-student SHAP explanations enable school counselors to design individualized interventions—for instance, a student whose dropout risk is primarily driven by poor behavior (F18) and low grades (F20) can be referred to psychosocial support and academic tutoring, respectively. At the institutional level, school directors can use the aggregate risk

TABLE 15. Usability Survey Results ($n = 10$). Scores on a Likert scale 1–5; mean ($\bar{x}$) and standard deviation (SD) are reported per question.

Dim.	Q	U0	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10	$\bar{x}$	SD
Usability 3.53	Q1	3	4	5	4	3	3	3	4	3	4		3.60	0.70
	Q2	4	4	3	5	3	3	3	3	5	4		3.70	0.82
	Q3	3	3	3	3	3	3	3	4	4	3		3.20	0.42
	Q4	4	4	5	3	3	3	3	3	5	3		3.60	0.84
Utility 4.00	Q5	4	4	4	5	3	4	4	3	4	4		3.90	0.57
	Q6	4	4	5	4	3	4	4	5	3	5		4.10	0.74
Satisf. 3.45	Q7	4	4	3	4	3	3	3	4	4	3		3.50	0.53
	Q8	3	4	3	3	3	4	4	3	4	3		3.40	0.52
	Q9	3	4	3	3	3	4	4	3	4	4		3.50	0.53
	Q10	3	3	3	3	4	4	4	4	3	3		3.40	0.52

rankings to prioritize resource allocation toward high-risk cohorts. At the policy level, the consistent identification of behavioral and academic factors as dominant predictors suggests that Peruvian dropout prevention programs should integrate behavioral monitoring alongside academic tracking. The system's deployability via a web interface—without requiring data science expertise from the user—is a critical enabler for adoption in resource-constrained schools.

EduPredict was tested by a group of $n = 10$ professionals from the education sector, including directors, psychologists, and teachers, to evaluate usability, utility, and satisfaction, through a questionnaire with alternatives according to the Likert scale (1–5). Considering that the average ratings are not integers, continuous intervals commonly used in various areas [38], [52], [53] were adopted as follows: 1.0–1.8: None; 1.81–2.6: Little; 2.61–3.4: Medium; 3.41–4.2: High; 4.21–5: Very High. The dimension-level mean scores were utility: 4.00, usability: 3.53, and satisfaction: 3.45, placing all three within the “High” interval (3.41–4.2) of the adopted Likert interpretation scale. However, given the low Cronbach's α values ($\alpha = 0.05$ –0.42) and the small sample size ($n = 10$), these figures should be interpreted as descriptive summaries of participants' individual ratings rather than psychometrically validated scale scores. No claim of internal-consistency reliability is made for the sub-scales. In the utility dimension, the identification of at-risk students (Q6: $\bar{x} = 4.1$, SD = 0.74) stood out, as did supporting educational decisions (Q5: $\bar{x} = 3.9$, SD = 0.57). In usability, ease of access to functions (Q2: $\bar{x} = 3.7$, SD = 0.82) and intuitive design (Q1: $\bar{x} = 3.6$, SD = 0.70) were highlighted. Internal consistency was assessed via Cronbach's alpha; computed values were $\alpha = 0.42$ for the usability sub-scale (4 items), $\alpha = 0.05$ for utility (2 items), and $\alpha = 0.21$ for satisfaction (4 items). These low values are consistent with

the multi-dimensional nature of the instrument and the small sample size ($n = 10$), and should be interpreted accordingly.

Limitations. Several limitations of this study must be acknowledged explicitly:

- **Dataset scope:** The dataset was collected from a single public secondary school in a peri-urban area of Metropolitan Lima. While this context is representative of schools in similar socioeconomic settings [54], results may not generalize to schools with markedly different profiles (e.g., elite private schools, highly rural schools with limited infrastructure).
- **Usability evaluation:** The usability assessment involved only $n = 10$ participants from the same institution. This sample is insufficient for statistically robust conclusions about system acceptability, and participants' familiarity with the institutional context may introduce positive bias.
- **Reliability of the evaluation instrument:** The low Cronbach's alpha values obtained are partly attributable to the small sample and the multi-dimensional nature of the questionnaire. A larger, multi-site evaluation would be required for more robust reliability estimates.
- **Single decision threshold:** The performance metrics are reported for a single classification threshold; a full ROC curve analysis across multiple thresholds would provide a more complete picture of model discrimination ability.
- **Temporal validation:** The study uses a single cohort snapshot. Longitudinal validation—training on historical data and predicting for future cohorts—is needed to confirm the model's predictive validity over time.

VII. CONCLUSION

This work presented EduPredict, a web application that integrates machine learning prediction with SHAP-based explainable artificial intelligence to address school dropout in public secondary schools in Metropolitan Lima, Peru. Five ML algorithms were implemented and compared on a dataset of 1,456 student records, with Gradient Boosting achieving the highest accuracy (97.27%) and precision (99.28%). The explainability analysis consistently identified student behavior and academic grades as the dominant dropout risk factors, followed by family structure and health-related variables—findings that align with the broader literature while highlighting locally relevant dynamics.

Unlike previous studies that treat prediction and explainability as separate research goals, EduPredict integrates them into a single, educator-facing tool that translates probabilistic predictions into actionable, student-specific explanations. This distinction is practically significant: educators are not merely told that a student is “at risk,” but are shown which specific factors drive that risk and by how much, enabling targeted and personalized interventions.

The descriptive evaluation by 10 educational professionals yielded mean ratings in the “High” range for utility (4.00/5), usability (3.53/5), and satisfaction (3.45/5), suggesting

practical acceptance of the system within a real institutional context, although the low internal-consistency estimates ($\alpha = 0.05\text{--}0.42$) preclude treating these as validated scale scores. At the same time, the identified limitations—single-school dataset, small usability sample, and single-threshold performance reporting—define a clear agenda for future work.

Future research directions include: (1) extending EduPredict to a representative sample of schools across Metropolitan Lima, covering diverse socioeconomic and geographic profiles; (2) applying clustering techniques (e.g., K-means) to develop specialized predictive sub-models for student subpopulations with distinct risk profiles [55]; (3) integrating EduPredict with the Peruvian national educational information system (SIAGIE) to enable automated, large-scale data ingestion and model retraining; (4) conducting longitudinal validation by training models on historical data and testing on future student cohorts; and (5) expanding the usability evaluation to a larger, multi-institutional sample to obtain statistically robust reliability estimates of the instrument.

AUTHOR CONTRIBUTIONS

Conceptualization, Andreé Barboza, Jhonatan Huamani, and David Mauricio; methodology, Jhonatan Huamani and David Mauricio; software, Andreé Barboza and Jhonatan Huamani; validation, Andreé Barboza, Jhonatan Huamani, José Santisteban, Nelson Maculan, and David Mauricio; formal analysis, Andreé Barboza, Jhonatan Huamani, and David Mauricio; investigation, Andreé Barboza, Jhonatan Huamani, and David Mauricio; data curation, Andreé Barboza and Jhonatan Huamani; writing—original draft preparation, Andreé Barboza and Jhonatan Huamani; writing—review and editing, Andreé Barboza, Jhonatan Huamani and David Mauricio; supervision, José Santisteban and David Mauricio; project administration, David Mauricio. All authors have read and agreed to the published version of the manuscript.

INFORMED CONSENT STATEMENT

Not applicable.

DATA AVAILABILITY STATEMENT

The data presented in this study are available on request from the corresponding author.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

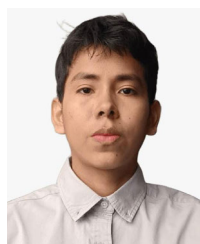
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